

STL: Orchestrating a global network with data analytics and AI to achieve process excellence

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ABOUT STL



About STL

STL built a data lake on Google Cloud to help employees easily access and visualize data to improve process efficiency and business decisions with the help of analytics and machine learning.

STL (<https://www.stl.tech/>) is an industry-leading integrator of digital networks with 25 years of experience in optical fiber innovations and India-made secure 5G solutions. Its core capabilities include optical interconnect, virtualized access solutions, network software and system integration. It enables full value connectivity by building end-to-end technology solutions for global networks.

Industries: Technology

Google Cloud results

- Enables predictive view of key business metrics by making STL focus on the highly correlated inputs metrics in the KPI tree
- Acutely helps with decision support to test complex hypothesis in high-precision manufacturing, directly impacting yield across global plants
- Drives right focus by guiding the Data Networks project teams on where to focus by proactively working on the

Generate insights with

Location: India

enterprise/

Cloud IAM (https://cloud.google.com/iam/)



About CloudCover

Founded in 2015, CloudCover is an award-winning cloud service provider specializing in infrastructure and data migration. It is one of the first Google Cloud partners in Southeast Asia and 2017 Google Cloud APAC Services Partner of the Year. Headquartered in India, it has satellite sales and project management offices in Singapore and Los Angeles.

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

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Dataflow (<https://cloud.google.com/dataflow/>)

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

Cloud IAM (<https://cloud.google.com/iam/>)

APAC Services Partner of the Year.

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace

(<https://workspace.google.com/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

enablers and
the detractors
for timely
completion of
global projects
in the field

- Improves team
collaboration
for 3,100
employees
working
remotely
during COVID-
19 pandemic
- Pub/Sub helps
to collect and
stream IoT
data from IoT
gateways to
Bigtable for
analysis in
milliseconds

An explosion in data is driving unprecedented demand for optical fiber, the specialist material that forms the internet's backbone. However, the fiber optics market is increasingly commoditized, and suppliers compete on price. To spark growth and innovation, optical fiber company STL (<https://www.stl.tech/>) has transformed itself into a data-driven digital business.

Innovation is in STL's corporate DNA. The company holds 358 patents, and its customer base includes

two of the world's top cloud companies and several tier-one telcos. Providing both the hardware and know-how to build, design, and manage next-gen networks, STL expects its market to grow from \$20 billion in FY17 to \$75 billion by FY23

(https://www.stl.tech/press_release/STL-delivers-superlative-growth-with-a-strong-future-outlook/)

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—Anand Agarwal, Group
CEO, STL

For STL to serve its customers efficiently, it needs to orchestrate its global network, from the flow of raw materials and order taking, to network design and deployment of optical fiber cables. As part of its transformation journey, STL built a data lake on Google Cloud (<https://cloud.google.com/>) to hold business process data and run AI/ML models for predictive analysis to support decision-making.

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Nischal Gupta, Chief Transformation Officer at STL.

“STL’s data lake, built in Google Cloud, empowers us to do that by securely managing and analyzing large volumes of diverse business data. It has also enabled us to build an integrated sales and marketing engine, aligned with our go-to-market strategy.”

Google Cloud offers STL an end-to-end solution, from smart analytics to workplace productivity. For its data lake, STL uses [Dataflow](#) (/dataflow) to extract, transform, and load (ETL) data from different datasets such as enterprise resource planning and customer relationship management into [BigQuery](#) (/bigquery). For edge IoT projects, STL uses [Pub/Sub](#) (/pubsub) for messaging between IoT gateways and Google Cloud and [Cloud Bigtable](#) (/bigtable) that’s optimized for high volumes of data. STL also deployed an AI chatbot using [Chat](#) (<https://workspace.google.com/products/chat/>), part of [Google Workspace](#) (<https://workspace.google.com/>), and [Dialogflow](#) (/dialogflow), part of [Contact Center AI](#) (/solutions/contact-center), to reduce the time and effort employees spend on routine tasks.

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performance
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—Nischal Gupta, Chief
Transformation Officer,
STL

Leading change with data-driven decision-making

So what can other companies learn from STL's
successful transformation journey?

"STL's transformation is based on robust foundations and scalability—enabling us to aim as high as possible! Our culture of giving ownership of data to our people, making decisions based on data rather than instinct, and building collaborative partnerships with business teams positioned us to seize value from machine learning and other new technologies," says Nischal.

STL's transformation is focused on solving business problems such as, "How can we produce at desired quality levels without delays? How do we automate low-value business processes? What are the innovative solutions that lend an edge to all customers? How do we deliver value for money to our customers?"

Data held the key to these answers. "Our business is all about data networks. With the help of BigQuery, we can use data internally to make the right decisions or even to make timely surgical interventions, be it around the efficiency of our manufacturing processes or the predictive models for our quantitative and qualitative performance metrics," says Nischal.

Executing a successful data lake initiative

Moving compute and storage capacity from on-premises hardware to Google Cloud was a first for STL. It called for careful planning and execution from a cross-functional team.

Internal stakeholders from data science, security, and engineering made up STL's evaluation team. The team decided what goes into the data lake, what processes to automate, what tools to select, how the solution design should look like, whether to outsource to a partner, and so on.

“Our architectural landscape and transformation design principles enabled us to avoid running separate data warehouses. Creating a data lake enabled us to extract deep insights from historical and real-time data from disparate sources in a unified, clean format. This is a

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—*Manuj Desai, Head of IT
Transformation, STL*

Apart from vendor presentations and interviews, STL conducted a six-month long proof of concept lab to test use cases in a real-world environment. Key considerations included technical solutions, partner support, security, the total cost of ownership, long-term partnership, and platform reliability.

“Google Cloud stood out for its ability to create BigQuery data structures that reflect our business flow so that business managers can easily make sense of the data,” says Jayakkumar Krishnasamy, Head of Data Engineering at STL. “Our good business judgement and basic analytical skills have now been complemented with deeper data insights to ensure that we are focusing on the right drivers of our business performance.”

According to Jayakkumar, BigQuery is able to run queries and generate results within five seconds, irrespective of the data size.

The horizontal architecture of STL's Google Cloud data lake cuts across applications, geographies, business functions, and critical business outcomes for supplier, production, quality, and customer

matters. It links all data flows, from sales opportunities to closures, and includes information around all contracts, won or lost.

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Google Cloud Partner [CloudCover](https://cldcvr.com/) (<https://cldcvr.com/>) took six months to complete STL's first data lake use case in June 2018 for processing machine IoT data. The five terabyte data lake continues to grow as the company creates more data each day.

“CloudCover is the secret sauce for our successful data lake implementation. Building a data lake for a company like ours is new to all of us,” says Jayakkumar. “CloudCover understood our vision and translated our business objectives into specific technical solutions on Google Cloud.” He adds that beyond this project, STL continues to work with CloudCover on new and complex Google Cloud projects.

Providing room to focus on new business development

The nature of STL's business requires it to consistently churn out data, and with data-lake-based computing, it is able to do this much more efficiently and timely. There was no consistent way to access data sources across different departments and business units. Data analysts tracked pipelines and ordered forecasts on spreadsheets for their respective business units.

Jayakkumar says that the data lake gives STL a holistic view of its solutions, operating geographies, and customers. At the same time, as a managed service, BigQuery allows the engineering team to focus on new development, rather than provisioning and maintaining infrastructure.

"Automating ETL pipelines on BigQuery means that data analysts don't have to spend time on data collection," says Manuj. "They can focus on analyzing and sharing their insights instead."

"STL also democratizes data access with DataDesk, an information portal that it hosts on [Sites](https://workspace.google.com/products/sites/) (<https://workspace.google.com/products/sites/>) to help employees find what datasets are available and who the data owner is. If someone has a business case for analytics, they share it with the data owner or the business owner for access to the relevant sources," says Mohit Mathur, Head of Data Science at STL.

Augmenting human intelligence with machine learning

STL is a process-focused organization that generates a vast amount of data. For example, the production process for optic fiber produces several gigabytes of data per day per machine. Processes can determine business profitability, whether through productivity improvements at a manufacturing plant or in timely installations of network services. However, the company lacked an infrastructure to ingest large amounts of disparate data for analytics.

Previously, STL utilized programmable logic controller (PLC) sensors mounted on 45 machines across seven factories to collect production equipment data such as shop floor efficiency and machine downtime. STL wanted to extract deeper insights by applying ML on production data on the data lake.

To transform its manufacturing data, STL created its own IoT gateways to capture complex parameters such as ratio of gases, soot deposition, and rotation speed of mandrel for drawing fiber. The company uses Pub/Sub to collect and stream the IoT data from the IoT gateways to Bigtable for analysis in milliseconds.

“BigQuery overcame our misconceptions around the idea that data could be too voluminous or too minute for analysis, or that analyzing it would take a long time,” says Manuj. “We managed to build ML models in just months instead of years, to access granular data from business processes. This enabled us to augment human intelligence by

providing the process engineering and R&D teams with insights that were not previously known.”

The ML models, running on Google Cloud and STL’s data lake, analyze real-time data and historical quality outcomes to deliver real-time recommendations. It helps STL predict future outcomes of current choices and therefore make optimal decisions. It has also sped up data-based decision-making by 3x over the past year.

“It was a major accomplishment to take AI/ML from an ‘ivory tower’ technology to a tool that’s touched and felt by everyone in the organization,” says Manuj. “The data lake, powered by Google Cloud, changed the way people perceived data. Today, there’s a clear understanding that data is an enterprise asset and an investment.”

“In our Lead 360 Solution offering, our complex projects are subject to many variables. We deployed a custom-built AI tool powered by BigQuery to predict the impact on project delays and margins from real-time data,” says Nischal, adding that by pinpointing exact areas of project intervention to stem delays, it can deliver projects on time and within budget to customers.

Improving employee interactions through self-service

Apart from machine learning to improve business outcomes, STL recognized the potential of using AI to resolve the bulk of employee queries.

STL built Mitra (meaning “friend” in Hindi), a self-service chatbot on Chat with Dialogflow to improve productivity and employee experience. Beyond answers to FAQ responses, Mitra helps employees perform a variety of routine tasks from attendance management to password resets, as well as escalating customers to a support staff, if it is unable to handle a specific request.

Since Google Cloud supports Single Sign-On (SSO) across all applications and services in STL's landscape, Mitra uses Cloud Identity (/identity) to deliver personalized content to employees. Time reporting, also known as attendance regularization, is a norm for manufacturing operations where employees are paid for each work day. STL's employees used to take 10 steps to regularize each absent day in the HR management system (HRMS). Now, they can now focus on more productive work.

Thanks to Mitra, employees can type in related keywords such as "attendance regularization" or "regularize," and the bot will automatically gather attendance data from HRMS and display the absent days in the chat. The employee can select reasons such as customer meeting or plant duty to regularize attendance for one or all of the dates. To date, employees have saved 7,500 hours on attendance marking by using Mitra on Google Chat.

STL also saves 700 hours a year on password resets with the help of Mitra. Like many organizations, password resets in STL account for the highest number of IT helpdesk tickets. Each request takes an average of five minutes to resolve. Instead of

contacting IT, an employee can type in password reset along with the user's name on Mitra. Mitra retrieves the credentials of the requestor on Google Cloud and initiates the reset process if the requestor is authorized to reset other user's passwords.

"Self-service is an important component of STL's transformation roadmap. Our transformation goal is to give employees access to information and tools so that they can focus on their core tasks," says Ajay. "For example, HR staff can spend time on talent retention and workforce planning, instead of answering standard questions about insurance coverage."

"The user-friendly chatbot greatly improves employees' satisfaction on internal processes," says Manuj. "That's reflected by internal feedback from 98% of employees who find Mitra extremely helpful and easy to use. We plan to improve Mitra with more user-friendly features such as speech-to-text capabilities on Mitra as well as personalized recommendations."

According to Ajay, in the near future, Mitra will be able to recognize user behavior such as regular queries for PO at the end of the month and prompt the user if they want the PO before they even ask for it.

Securing product innovation with encryption and governance

For STL, security is of utmost importance because competitors could develop similar products if they

got hold of STL's intellectual property.

"Our data holds our story, our insights, and is a critical asset to our organization that we need to secure," says Manuj. "Google Cloud security features such as Cloud Identity and Access Management (/iam) (IAM) allows us to grant fine-grained access control so users can access the right amount of data without compromising data security."

STL needs to maintain data flow with the customers, partners, and suppliers in its global ecosystem without comprising cybersecurity. The Google Cloud security model for data encryption during transit and at rest helps to keep STL secure and compliant.

"From access management to the titanium chip for hardware encryption, the Google Cloud security model was a major factor in giving us confidence in the cloud platform," shares Manuj.

The COVID-19 pandemic also pressed the urgency for a secure workspace for STL developers working from home. The solution is a desktop as a service (DaaS) environment running on Google Kubernetes Engine (/kubernetes-engine). Authorized users can access the thin-client with Google Workspace credentials using Identity-Aware Proxy (/iap) on any computer from anywhere. This means STL can overcome COVID-19 movement restrictions by onboarding new joiners without hardware.

"Our DaaS environment using Google Kubernetes Engine provides remote developers with the right amount of security, without slowing down the development cycle," says Ajay. "To protect IP, these

non-intrusive security controls prevent users from copying code from the dev environment to any other application, be it a local laptop, browser, note, or email apps.”

According to Ajay, each DaaS environment can be further optimized for the number of compute cores and memory. “For example, an engineering software developer may need eight core, 16 GB, while an IT developer may need only four core, eight GB. I can fine-tune each environment based on the role or department that the person works in.”

Breaking down geographical barriers through collaboration

More than 3,000 employees work at STL across 16 countries. Shifting from a product-focused model to an end-to-end services model requires people and teams to collaborate and deliver value to a common set of customers. When STL rolled out Google Workspace (then G Suite) in 2016, it removed business silos and transformed the way people work as “One STL.”

“Google Workspace introduced us to a collaborative work style that's now ingrained throughout the organization. This is a far cry from when we needed the help of IT just to talk to each other,” says Nischal. Before Google Workspace, the team required five IT personnel to set up a video conference call between business units. “Now when we host town halls on Meet, everyone knows how to dial in from Calendar from any device without calling IT for help.”

STL held education sessions to help employees learn how Google Workspace can improve the way they work. In the beginning, some employees didn't know how to click on a link to open up an online document, let alone raise a comment or resolve the comment.

"Collaboration now happens so seamlessly. Instead of sending an attachment via email and waiting five to 10 days for a response, employees are using Docs (<https://workspace.google.com/products/docs/>) to work faster by collaborating on documents," adds Nischal.

Employees at STL score culture shifts and collaborative workforce very highly. "Employees tell us that the company's timely adoption of a cloud-first strategy and Google Workspace during COVID-19 has been beneficial, because it allows them to work remotely without disruption. As a result, they've been able to work more efficiently than ever," says Nischal.

Moving forward through process innovation

STL is currently working on a suite of ML models that provide demand forecasts to support financial planning and investor relations, among other initiatives.

"Transformation is an ongoing process," says Nischal. "We want to build an intelligent organization for now and the future. Success in this area will depend not only on intelligent people but on

intelligent processes, and Google Cloud delivers a best-in-class and an agile cloud platform to help us move forward.”